From Reconstruction to Anticipation

Predicting future tissue geometry from monocular laparoscopic video

FYP-2 Mid Evaluation · Modules 5 and 6

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What the project is for

Reconstruct a surgical scene in 3D from one laparoscope camera, then anticipate how the tissue deforms next — with no depth sensor and no clinical annotation.

O1 · Reconstruct
4D scene geometry from monocular video.

COMPLETE · FYP-1

O2 · Learn deformation
Self-supervised model of tissue motion.

COMPLETE · MODULE 5

O3 · Predict ahead
Future geometry at 1–3 frames, with confidence.

IMPLEMENTED · MODULE 6

O4 · Evaluate
Against baselines and the literature.

IN PROGRESS · M7–M8

Research question

Can future 3D tissue geometry be predicted from monocular surgical video alone — and can the system say how much to trust each prediction?

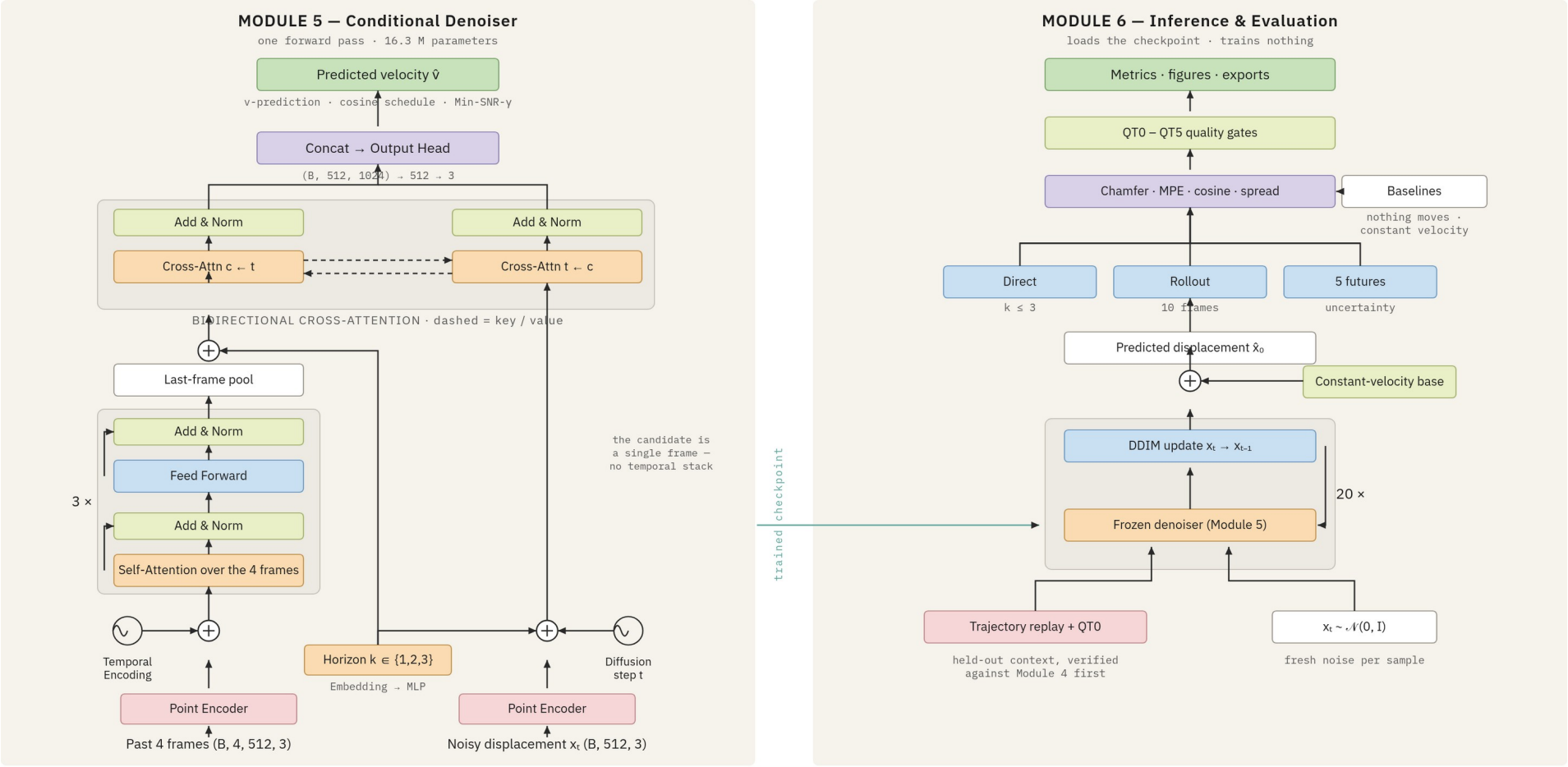
Faculty feedback from FYP-1, and what we did about it

Implement a base paper for comparative analysis	→	**Two base papers implemented** PointRNN (Fan & Yang) for point-cloud forecasting; Luo & Hu (CVPR 2021) for point diffusion. *module7_baselines.py*
Select a few SOTA papers and implement them	→	**Eleven methods on one protocol** Five classical and five learned predictors, plus ours, all scored on identical windows. *module7_baselines.py · 781 lines*
Do a detailed analysis — hyperparameter tuning, optimisation	→	**Four tuning studies** Inference cost against accuracy, context length, optimiser sensitivity, and diffusion ablations. *module8_hpo.py · 593 lines*
Start writing a paper in IEEE or Springer format	→	**IEEE conference manuscript drafted** Six pages, 22 references, tables generated directly from the result files. *paper/main.tex · paper/main.pdf*

All four are implemented and in the repository. Two still need their compute runs, which is item 1 on the remaining-work slide.

Architecture — Modules 5 and 6

Given the last 4 frames of 512 tracked points, predict their displacement 1–3 frames ahead — trained on 10 videos, no clinical labels.



Metrics chosen, and what they measured

The three questions we measure

Is the predicted shape in the right place? — Chamfer Distance
Is the motion pointing the right way? — directional cosine
Does the model know when it is unsure? — spread vs. error

We need the second one because a shape score alone cannot tell "learned nothing" from "right direction, wrong distance".

What the answers were

Accuracy — not yet ahead

CD 0.01568 vs naive 0.01557. Constant velocity beats both, on all 10 videos.

Direction — clearly learned

Cosine +0.57 at t+1. Nothing-learned would read 0.00. So: magnitude, not direction.

Uncertainty — calibrated

Spread vs realised error, r = 0.69. No baseline provides this at all.

Two explanations tested and eliminated before we concluded anything
No drift (ρ = +0.02, p = 0.61) · no dropped frames (333 origins checked, 0 found)

Six quality tests run as a suite. QT0 verifies the evaluation data is the data the model trained on, and gates everything downstream.

What we build on, and where we go past it

The work we build on		Where it stops, and what we add
EndoGaussian · Deform3DGS — surgical reconstruction Reconstruct a surgical scene in 3D at high fidelity, using a stereo camera.	→	**They only ever describe what has already happened.** We reach comparable fidelity from a single camera — and then predict forward from it.
PointRNN — point-cloud forecasting Predicts where a moving 3D point cloud will be next. Our closest published method, and our main learned baseline.	→	**Built for rigid outdoor scenes, and gives one answer with no confidence.** Same task on deforming soft tissue, and ours says how much to trust each answer.
Luo & Hu (CVPR 2021) — point-cloud diffusion The first diffusion model to generate 3D point clouds.	→	**It generates plausible shapes; it does not predict a future.** Our denoiser is that idea made conditional and temporal, so it forecasts rather than invents.
TPG-VAE · SurgVista — surgical anticipation Do predict the surgical future, and do it well.	→	**In 2D pixels, and only when told in advance what the instrument will do.** We predict 3D geometry, and we are not told the action.

No published method does all of it at once — predict the future, in 3D, on soft tissue, from a single camera, and say how confident it is.

Remaining work until the final evaluation

1. **Run the 11-method comparison and the HPO studies**
   Modules 7 and 8 are written and tested. Only the compute runs remain.
2. **Evaluate the two corrections already implemented**
   Magnitude calibration needs no retraining; the residual target needs one retrain.
3. **Move to leave-one-video-out evaluation**
   Removes the optimism of selecting and reporting on the same recordings.
4. **Fold the completed results into the manuscript**
   The IEEE draft is written with the tables generated from result files.
5. **Submit, and extend the corpus beyond 10 recordings**
   The overfitting we measured is consistent with a corpus this size.

DEMONSTRATION

What you are about to see, running live

Nothing in this demonstration is pre-rendered. The checkpoint is loaded and inference is run in front of you.

1

Load the trained checkpoint and verify the evaluation data
QT0 runs first: the rebuilt trajectory is checked against Module 4's saved windows before anything is predicted.

2

Predict a held-out moment at t+1, t+2 and t+3
DDIM sampling, 20 steps, against the naive and constant-velocity baselines on the same window.

3

Render the prediction as actual video frames
The predicted motion of 512 points is carried onto all 58,000 Gaussians and rasterised into an image.

4

Compare side by side with the real frame
Real · reconstruction ceiling · our prediction · naive baseline, with magnified motion arrows and an error map.

5

Forecast past the end of the recording
Twelve frames beyond the last recorded image — the only output that shows prediction rather than describing it.

Making the prediction visible

A prediction moves 512 points. A picture needs all 58,000. Bridging that gap is what turns a metric into something a surgeon could look at.

From points to pixels

Predicted motion of the 512 tracked points is carried onto every Gaussian by distance-weighted interpolation, so a tracked point receives its own motion exactly.

The renderer, re-verified

Module 2's renderer was re-implemented here, including degree-3 spherical harmonics. We checked it against the original rather than assuming they matched.

Six panels per moment

Real frame, reconstruction ceiling, prediction, naive baseline, motion overlay and an amplified error map — with image metrics over the full frame and over tissue alone.

Predicted motion was invisible

At 1–2 pixels, predicted and real frames look identical. Motion is drawn as arrows coloured by agreement with truth, magnified automatically — and the factor is printed on every panel, never applied to any reported number.

Nothing on screen was a prediction

Every predicted frame used to be paired with a frame that already existed. The forecast now runs 12 frames past the end of the recording: the real-video panel stops and says so, while the predicted panel keeps going.

30.04
reconstruction ceiling, dB

26.79
our prediction, dB

26.74
naive baseline, dB

The model knows which way tissue moves — not how far

Chamfer Distance alone cannot separate "learned nothing" from "learned the direction but the wrong distance". Measuring them separately can.

Agreement between predicted and true motion direction

t+1	t+2	t+3	t+5	t+10
+0.57	+0.40	+0.26	−0.01	−0.49

Why this matters

A model that learned nothing would read 0.00 everywhere.
+0.57 with naive-level error can only mean the magnitude is wrong — a far more tractable problem.

What follows from it

- Predict the correction to constant velocity (Module 5)
- Calibrate the magnitude at inference (Module 6)

Both implemented; evaluation is the next run.

The model knows when it should not be trusted

r = 0.69
spread vs actual error

5
futures sampled per input

5×
uncertainty growth over 10 frames

When the five sampled futures disagree, the prediction really is worse.

Uncertainty rises smoothly with how far ahead we predict — 0.00011 at one frame, 0.00055 at ten.

Why this is the result we are proudest of

A system that is confidently wrong is unusable in theatre.
A system that signals "do not trust me here" is usable even when imperfect.

No method in our comparison provides this.

Stated carefully

This is correlation between predicted spread and realised error. Some of it may reflect that busy moments are both harder and more variable — we say so in the paper rather than claiming more than we measured.

A two-line formula beats our model

Method	k = 1	k = 2	k = 3
Nothing moves (naive)	0.0093	0.0165	0.0222
Constant velocity	**0.0055**	**0.0138**	**0.0187**
Ours — diffusion	**0.0093**	**0.0165**	**0.0225**

Chamfer Distance, lower is better · full held-out set, 10 videos

Read this honestly

Our model sits exactly on the naive baseline at every horizon.

Constant velocity beats both — by 41% at one frame ahead.

It loses on all 10 videos.

Rollout stays stable (growth exponent 0.61) and two explanations were tested and eliminated:

- Error does not grow with distance from training frames (ρ = +0.02, p = 0.61)
- Dropped preprocessing frames do not explain it (333 origins checked, 0 drops found)